



Polystyrene microplastics facilitate renal fibrosis through accelerating tubular epithelial cell senescence

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ABSTRACT

Microplastics (MPs), emerging contaminants, are easily transported and enriched in the kidney, suggesting the kidney is susceptible to the toxicity of MPs. In this study, we explored the toxicity of MPs, including unmodified polystyrene (PS), negative-charged PS-SO₃H, and positive-charged PS-NH₂ MPs, in mice models for 28 days at a human equivalent concentration. The results showed MPs significantly increased levels of UREA, urea nitrogen (BUN), creatinine (CREA), and uric acid (UA) levels in serum and white blood cells, protein, and microalbumin in urine. In the kidney, MPs triggered persistent inflammation and renal fibrosis, which was caused by the increased senescence of tubular epithelial cells. Moreover, we identified the critical role of the Klotho/Wnt/ β -catenin signaling pathway in the process of MPs induced senescence of tubular epithelial cells, promoting the epithelial-mesenchymal transformation of epithelial cells. MPs supported the secretion of TGF- β 1 by senescent epithelial cells and induced the activation of renal fibroblasts. On the contrary, restoring the function of Klotho can alleviate the senescence of epithelial cells and reverse the activation of fibroblasts. Thus, our study revealed new evidence between MPs and renal fibrosis, and adds an important piece to the whole picture of the plastic pollution on people's health.

1. Introduction

Plastics are an integral part of our daily lives due to their properties including low price, robustness, durability, light weight, thermal insulation and resistance to degradation. It is estimated that plastic production in various regions of the world is about 350 million tons per year, of which 71% of plastic waste is discharged into terrestrial or aquatic environments (Bucknall, 2020; Busch et al., 2023). Under natural conditions, plastics undergo photooxidative and thermal oxidation degradation processes, physical wear, UV irradiation, and biodegradation effects, breaking down into microplastics (MPs) at 0.1 μ m to 5 mm (Wang et al., 2018). Humans are mainly exposed to MPs through dietary intake, air inhalation, or skin contact (Qu et al., 2023; S Wang et al., 2023). Growing evidence has demonstrated that MPs can be detected in human serum samples (Dusza et al., 2023), human urine (Pironti et al., 2022), and human stools (Kannan and Vimalkumar 2021), suggesting these refractory and small MPs are easily ingested and accumulated in

humans, causing toxic and side effects on multiple organs.

Chronic kidney disease (CKD) is a worldwide disorder with high incidence. It is caused by various disease pathways, leading to adverse changes in renal structure and function, and renal fibrosis is a common pathological process of CKD with various etiologies (Zhang et al., 2021). CKD of unknown etiology (CKDu) is prevalent worldwide (Gifford et al., 2017). A better understanding of the etiology of CKDu and finding therapeutic targets are conducive to the implementing of public health responses and health policies. Experimental studies showed chronic lead exposure has nephrotoxicity, causing the occurrence of CKD and renal failure (Goyer, 1989). Sub-chronic cadmium exposure induced renal interstitial fibrosis in rats with the Wnt/ β -catenin signaling (Liang et al., 2021). After exposure to herbicides in rats, massive inflammatory factors are enriched in the kidney and oxidative stress occurs, causing epithelial mesenchymal transformation (EMT) of epithelial cells in kidney tubules, promoting the occurrence of renal fibrosis (T Wang et al., 2023). More and more evidence indicated that environmental

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pollution may be an important factor in the occurrence of this CKDu.

Fibrotic renal injury consistently exhibits a variety of natural aging phenomena, such as interstitial fibrosis, loss of renal tubular cell repair capacity, and massive release of reactive oxygen species (ROS) and inflammatory factors (Verzola et al., 2008; Zhou et al., 2015). These observations remind us that renal fibrosis may be closely related to premature senescence (Stenvinkel and Larsson, 2013). Previous animal studies have demonstrated the enrichment of MPs in the renal tissues, causing mitochondrial dysfunction, endoplasmic reticulum stress, massive release of ROS, and inflammation in renal cells (Wang et al., 2021). Mitochondrial dysfunction, massive release of ROS and secretion of inflammatory factors are essential features of renal senescence (Lin and Epel, 2022). The above experimental results suggest that MPs exposure is likely to lead to early aging of certain cells in the kidney, followed by fibrosis and inflammatory microenvironment. Of note, growing evidence has pointed out that in the senescent model of the renal tissues, renal tubular cells are more easily to senescence, suggesting whether MPs cause renal fibrosis by accelerating the aging of renal tubular cells.

Thus, in this study, we used PS, PS-SO₃H, and PS-NH₂ MPs (5 µm) to explore whether the damage of epithelial cells in renal tubular induced by MPs might induce the occurrence of renal fibrosis. We further clarified the *in vivo* and *in vitro* experimental mechanisms of MPs-induced epithelial cell senescence in renal tubular. To more accurately assess the toxicity of MPs to humans, we exposed mice to equivalent concentrations of MPs in humans, providing new insights into the etiology of renal fibrosis.

2. Materials and methods

2.1. Materials and cell lines

The unmodified polystyrene (PS, 5 µm), amino-modified (PS-NH₂, 5 µm), and sulfonic-modified (PS-SO₃H, 5 µm) MPs were purchased from Tianjin Baseline ChromTech Research Center. ICG-001 (Wnt/β-catenin inhibitor) and rapamycin (Klotho agonist) were obtained from Medchemexpress (USA). Rat renal interstitial fibroblasts (NRK-49F cells) and epithelial cells in human proximal tubular (HK-2 cells) were obtained from ATCC. NRK-49F cells were incubated with DMEM medium with penicillin-streptomycin (1%) and FBS (10%), and HK-2 cells were cultured in HK-2 complete medium (CM-0109, Pricella, China).

2.2. Animal model

The BALB/C mice (three weeks, male) were obtained from the Yangzhou University Animal Center and were maintained in compliance with institutional policy. The mice were placed in the animal center room (20–22 °C). In this experiment, different charges of MPs were dissolved in distilled water, and mice were exposed to PS, PS-HSO₃ and PS-NH₂ MPs by oral gavage at an environmentally relevant concentration (0.5 mg/d) for 4 weeks according to the previous study (Pan et al., 2023). It has been previously reported that the daily intake of plastic particles in humans weighing 60 kg is about 0.23–11.9 mg/kg (Xu et al., 2023). The surface area is a crucial indicator for calculating the exposure dose conversion between mice and humans, and the calculation formula was: Mouse equivalent dose (mg/kg) = human dose (mg/kg) × [K_m (human)/K_m (mouse)]. Among them, K_m factor = weight (kg)/body surface area (m²). The body surface area and body weight of humans were 1.62 m² and 60 kg, respectively, and the indexes of mice were 0.007 m² and 0.02 kg, respectively (Nair and Jacob, 2016). According to the above parameters, the equivalent dose of 0.02 kg mouse was about 0.05–2.9 mg/day. In our experiment, we chose a daily dose of 0.5 mg MP. Serum biochemical parameters including UREA, uric acid (UA), creatinine (CREA) and urea nitrogen (BUN) were detected with the Cobas 8000 c702 module analyzer system. Urine parameters were detected by the Uritest-500B.

2.3. Histology and immunofluorescence

Fresh renal organs were fixed overnight with 4% PFA and embedded in paraffin. The embedded kidney was sectioned to 4 µm. Haematoxylin and eosin (H&E), sirius red, and Masson trichrome staining were conducted by standard protocol (Luo et al., 2018). Immunofluorescence staining was performed using routine protocol (Chen et al., 2017). The antibodies were listed as follows: anti-TNFα (Proteintech, China), anti-p21 (Santa, USA), anti-γH2AX (Proteintech, China), anti-Wnt4 (Abcam, USA), anti-TGF-β1 (Servicebio, China), and α-SMA (Proteintech, China).

2.4. SA-βgal staining

Cellular senescence in renal tissue and HK-2 cells were examined with β-Galactosidase senescence staining kit (Solarbio, China) (Pan et al., 2023). The protocol is based on the manufacturer's instructions. Images were collected via an inverted microscope, and Image J software was used to evaluate the number of senescent HK-2 cells.

2.5. Western blotting analysis

The renal tissues or HK-2 cells were wholly lysed in RIPA buffer containing protease and phosphatase inhibitors (MCE, China). Protein extracts from tissues and cells were separated by SDS-PAGE and transferred onto PVDF membranes. The membrane was blocked with 5% milk at room temperature for 1 h and incubated with primary antibody (IL-1β, Santa, USA; TNFα, Proteintech, China; p21, Santa, USA; γH2AX, Proteintech, China; Klotho, Proteintech, China; Wnt4, Proteintech, China; TGF-β1, Servicebio, China; E-cadherin, Servicebio, China; Vimentin, Servicebio, China; β-catenin, Servicebio, China; Fibronectin, Servicebio, China). The membrane was incubated with secondary antibodies, and the data was quantified with ImageJ software.

2.6. Quantitative real-time PCR (RT-PCR)

The total RNA of the renal organs and cells were incubated with TRIzol reagent (Vazyme, China). HiScript 1st strand cDNA Synthesis Kit (Vazyme, China) was used for RT-PCR reaction. cDNA was amplified in a mixture reaction system with SYBR Green Q-PCR Kits (Vazyme, China).

2.7. Statistical analysis

SPSS 18.0 was used for statistical analysis. All data were expressed as mean ± SD of at least three independent experiments. *P* < 0.05 was considered statistically significant. One-way ANOVA followed by Tukey's multiple comparison tests were used to analyze the differences between groups.

3. Results

3.1. MPs exposure led to abnormalities in renal function and urine routine

The experimental data showed that PS, PS-SO₃H, and PS-NH₂ MPs were spherical, uniform in size and smooth in surface. The mean diameters of PS, PS-SO₃H, and PS-NH₂ MPs were 4818 ± 307 nm, 4726 ± 442 nm, and 4894 ± 394 nm in water, respectively, which are close to the designed value as expected (Figure S1). The circularity of PS, PS-SO₃H and PS-NH₂ MPs were 0.979 ± 0.021, 0.977 ± 0.026, and 0.982 ± 0.012 in water, respectively (da Costa Araujo et al., 2020). The zeta potential of PS and PS-SO₃H MPs showed negative charge, while the zeta potential of PS-NH₂ MPs was opposite (Figure S1).

After MPs treatment, the body weight gain was markedly inhibited relative to the normal group (Fig. 1A). However, after 28 days of treatment, we failed to observe the differences in the paired kidney index. The serum UREA and BUN were significantly higher in PS-NH₂

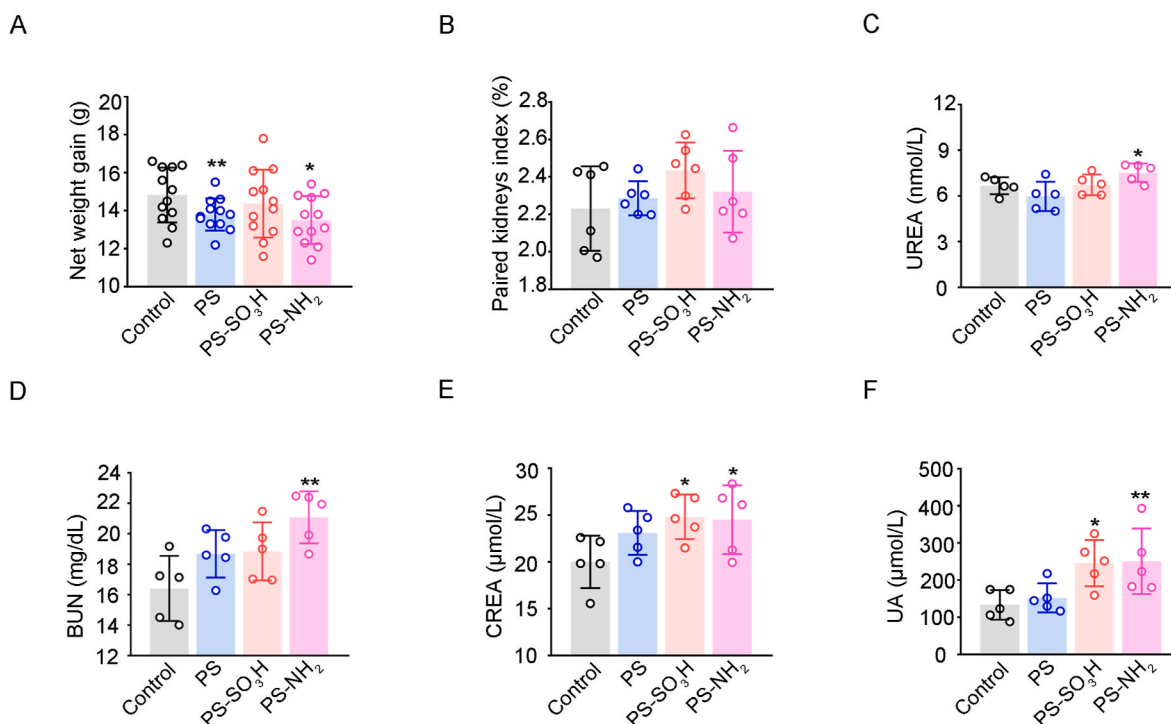


Fig. 1. MPs exposure caused abnormalities in renal routine. The mice were given ddH₂O and three groups of MPs by gavage every day for 4 weeks. (n = 12) (A, B) Body weight gain and paired kidney index (%) were recorded during the period. (C–F) Blood biochemistry was analyzed via assessment of UREA, creatinine (CREA), uric acid (UA), and urea nitrogen (BUN) (n = 5). Data were presented as means ± SD. P* < 0.05, P** < 0.01.

MPs group, while no significant change was observed in PS and PS-HSO₃ MPs groups. Meanwhile, PS-HSO₃ and PS-NH₂ MPs treatment increased the CREA and UA levels. In the routine urine test, our data demonstrated that white blood cells (WBC), protein (PRO), and microalbumin (MA) are significantly increased (Table S1), suggesting that inflammatory and structure injury may occur in the urinary system.

3.2. MPs exposure induced kidney fibrosis

The groups treated with MPs exhibited increased the collagen fibers in the kidney tubule and kidney medulla, suggesting toxicity of MPs damage the kidney (Fig. 2A–D). Moreover, the mRNA levels of fibrotic indicators (Col1a1, α-SMA, Col3a1, and fibronectin (FN)) were effectively elevated following MPs treatment in relation to the control group (Fig. 2E). The above results indicated that MPs exposure induces kidney fibrosis.

3.3. MPs exposure accelerated renal tubular epithelial cells senescence

Cellular senescence is related to the progression of renal diseases, and accelerating tubule cell senescence induces renal fibrosis (Luo et al., 2018). We first assessed SA-βGal staining in renal tissues to detect the senescent cells, and three groups of MPs exposure caused a significant increase in SA-βGal positive staining of epithelial cells in renal tubular (Fig. 3A and B). Consistently, the kidney exposed to MPs revealed multiple senescent phenotypes, such as enhanced secretion of IL-1β and TNFα, and significantly increased levels of p21 and histone γH2AX. γH2AX is a selective marker of DNA double-strand breaks, which is considered as an important feature of senescence (Cardus et al., 2013). Furthermore, we evaluated the protein level of Klotho, a marker of tubular injury (Zhou et al., 2013), whose expression was significantly suppressed (Fig. 3C, D, Figure S2). Similar results were found in the immunofluorescence staining of renal sections with antibodies against TNFα (Fig. 3D, Figure S2) and p21 (Fig. 3E and F), and fluorescence expression mainly persisted in the kidney tubule. We next examined the

upregulated expression of vimentin in MPs-treated kidneys, and significant decreased changes in E-cadherin were observed (Fig. 3G and H).

3.4. MPs exposure induced cellular senescence in renal tubular epithelial cells in vitro

To assess the cytotoxicity of MPs on tubular cells, we used MTT assay to evaluate the viability of HK-2 cells exposed to MPs. With PS and PS-HSO₃ MPs (240 μg/mL) treatment for 48 h, the cell viability was significantly impaired (Fig. 4A and B). For PS-NH₂ MPs, cell viability decreased at 60 μg/mL or higher concentrations (Fig. 4C). For PS-HSO₃ MPs, cell viability decreased by 96 h at 60 μg/mL to 240 μg/mL (Fig. 4B). In addition, significant cytotoxicity was observed in HK-2 cells after exposure to 30 μg/mL or higher doses of PS-NH₂ MPs for 96 h (Fig. 4C). Regarding potential charge, the cytotoxicity of positively charged PS-NH₂ was significantly higher than that of other compounds. Considering that humans are most likely to be exposed to low levels of environmental MPs, we finally selected 60 μg/mL for subsequent research. In cultured HK-2 cells, incubation with MPs triggered the upregulation of SA-βGal activity (Fig. 4D and E). Consistent with these results, the increased levels of senescence-associated proteins including IL-1β, TNFα, p21 and histone γH2AX, and the decreased expression of Klotho were found (Fig. 4F, Figure S3). MPs treatment markedly induced an increase in γH2AX nuclear foci in HK-2 cells (Fig. 4G, Figure S3). As predicted, we found that E-cadherin protein levels were markedly reduced, while vimentin protein levels were increased in response to MPs treatment (Fig. 4H and I).

3.5. Identification of the responsible Wnt in senescent epithelial cells in renal tubular

The Wnt/β-catenin pathway is kept in silent in physical conditions of the kidneys, while activated in the process to promote renal fibrosis (Miao et al., 2019). To confirm the role of Wnt family members that worked in kidney fibrosis, we comprehensively analyzed the mRNA

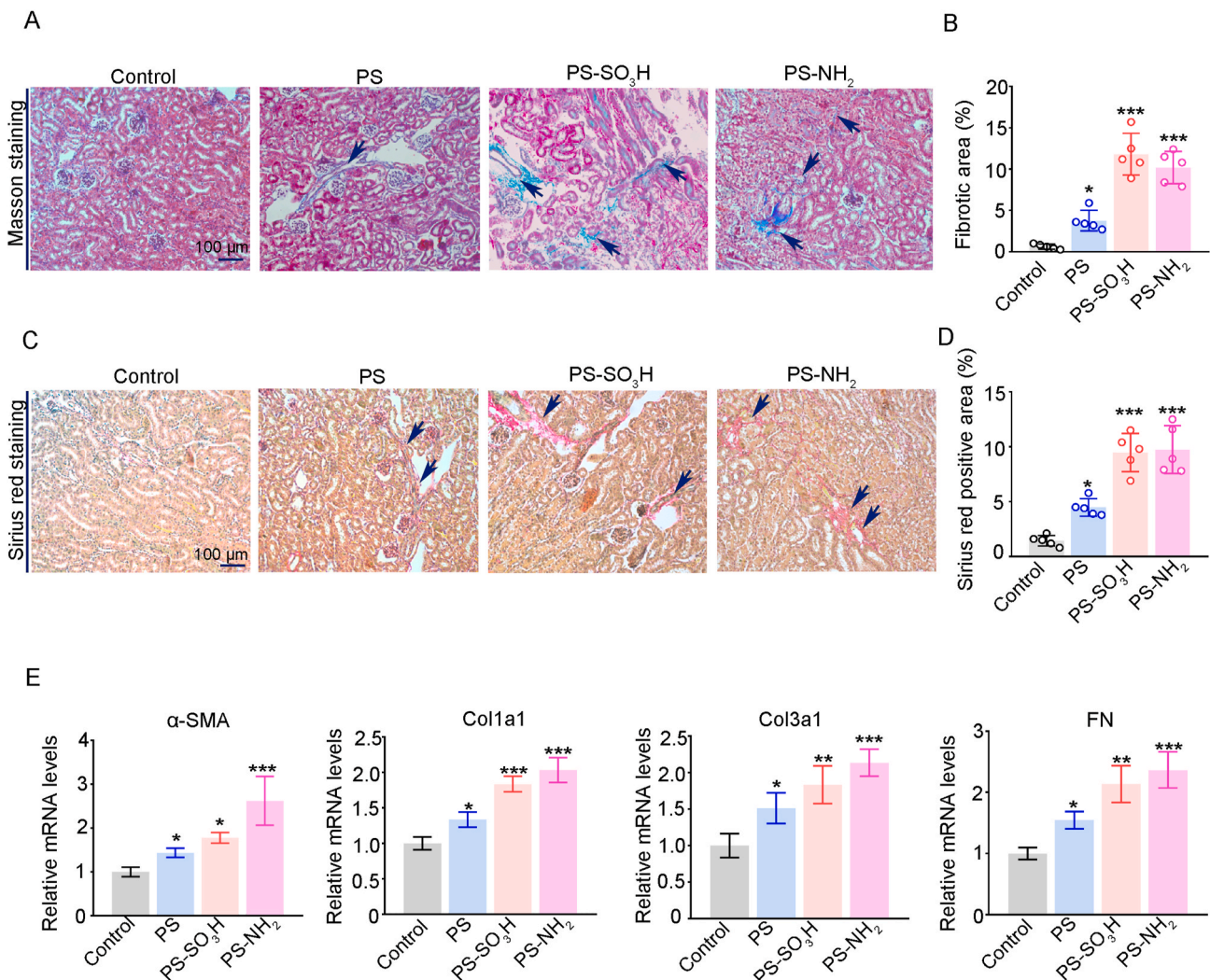


Fig. 2. MPs exposure induced kidney fibrosis. (A–D) The representative micrographs suggest collagen deposition, as indicated by Masson trichrome staining and Sirius red staining. (E) The mRNA expression of fibrosis-related genes was detected by qPCR analysis (n = 5). Data were presented as means ± SD. P* < 0.05, P** < 0.01, P*** < 0.001. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

expression of 19 Wnt genes in mice and HK-2 cells exposed to MPs. Among Wnt genes, Wnt4 was especially overexpressed *in vivo* and *in vitro* (Fig. 5A and B). In Fig. 5C, Wnt4 genes were markedly increased in both cells and fibrotic mouse tissues. MPs promoted the activation of Wnt/β-catenin signal pathway with inducing the expression of β-catenin (Fig. 5C and D). Moreover, we found that MPs induced mRNA levels of Snail 1, FN, and MMP-7, which are key transcription factors for β-catenin signaling and are considered the 'master genes' controlling renal fibrosis (Fig. 5E).

3.6. Klotho mediated Wnt/β-catenin signaling in senescent renal tubular epithelial cells

Administration of ICG-001 (Wnt/β-catenin inhibitor) markedly suppressed MPs-induced vimentin levels and reversed E-cadherin levels in HK-2 cells (Fig. 6A–D). Immunolabeling also exhibited that MPs profoundly up-regulated the expression of Wnt4 and inhibited the anti-aging protein Klotho (Fig. 6E). Furthermore, rapamycin (Klotho agonist) exerted inhibitory effects on the cellular senescence of HK-2 cells *in vitro* (Fig. 6F). In order to further verify the function of Klotho in Wnt/β-catenin pathway, we also examined the protein expression of β-catenin in cells, and reduced levels in β-catenin with rapamycin treatment (Fig. 6G). Our data suggested that the Klotho/Wnt/β-catenin pathway is

pivotal in MPs-induced renal fibrosis.

3.7. Senescent epithelial cells in renal tubular secreted TGF-β1 to activate fibroblasts

Immunofluorescence staining showed that three kinds of MPs treatment significantly increased TGF-β1 expression in tubules and HK-2 cells (Fig. 7A and B). Based on the concept that MPs treatment may result in a renal fibrosis phenotype in mice, we evaluated the effect of MPs exposure on renal cell-cell communication. HK-2 cells were pretreated with PS, PS-SO₃H, and PS-NH₂ MPs for 96 h to induce cellular senescence. Then, the supernatant was collected from HK-2 cells and used to stimulate NRK-49F cells (Fig. 7C). Compared to the control groups, NRK-49F cells incubated with supernatant from HK-2 cells treated with MPs exhibited elevations of α-SMA, Col1a1, and Col3a1 (Fig. 7D–F). Next, we explored the function of cellular senescence in MPs-induced kidney fibrosis. To investigate this, HK-2 cells were incubated with rapamycin and then exposed to different charges of MPs. Then, we collected the supernatant of HK-2 cells, and the medium was used to culture NRK-49F cells, and administration of rapamycin markedly suppressed MPs-induced fibronectin levels in NRK-49F cells (Fig. 7G). As predicted, the fibrosis indexes were significantly inhibited (Figure S4).

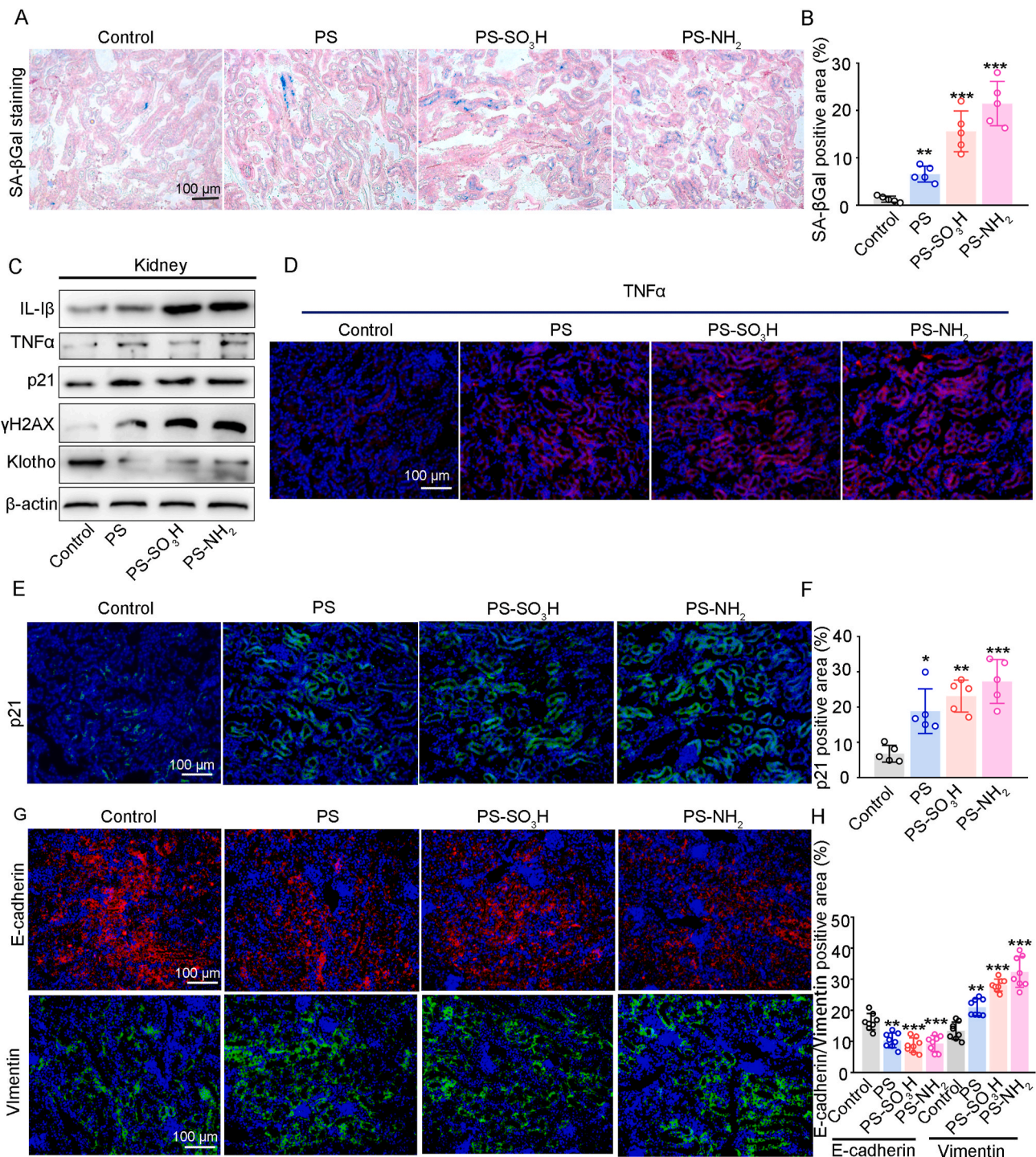


Fig. 3. MPs exposure accelerated renal tubular epithelial cells senescence. (A, B) The SA-βGal staining images (blue) were calculated using Image J software to quantify the SA-βGal positive area ($n = 5$). (C) The protein levels of IL-1β, TNFα, p21, γH2AX, and Klotho in kidney were determined by western blotting. (D–F) The content of TNFα and p21 in kidney were detected using immunofluorescence assay, and quantifications of p21 positive area was calculated by Image J software ($n = 5$). (G, H) The content of E-cadherin and vimentin in kidney were determined by immunohistochemical method, and the quantifications of vimentin and E-cadherin positive area were calculated using Image J software ($n = 8$). Data were presented as means $\pm$ SD. $P^* < 0.05$, $P^{**} < 0.01$, $P^{***} < 0.001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

4. Discussion

Recent studies have demonstrated that MPs can penetrate multiple biological barriers, enter the circulatory system, and produce toxicological effects on different organisms, especially for liver, reproductive and nervous system (Lu et al., 2016; Sussarellu et al., 2016; Yu et al., 2018). Nevertheless, understanding about the biological effects of MPs

on kidney damage remains limit. The renal tissues are the pivotal organ responsible for the metabolism, transportation and excretion of metabolic waste (Ruan et al., 2021). Growing evidence in human beings and animals have indicate that urine is a metabolic pathway for MPs (Pironti et al., 2022; Tsou et al., 2023; Wu et al., 2022). The kidney is an organ that forms urine. Once the function of the renal organs is abnormal, the urine will also change accordingly. The current study established a

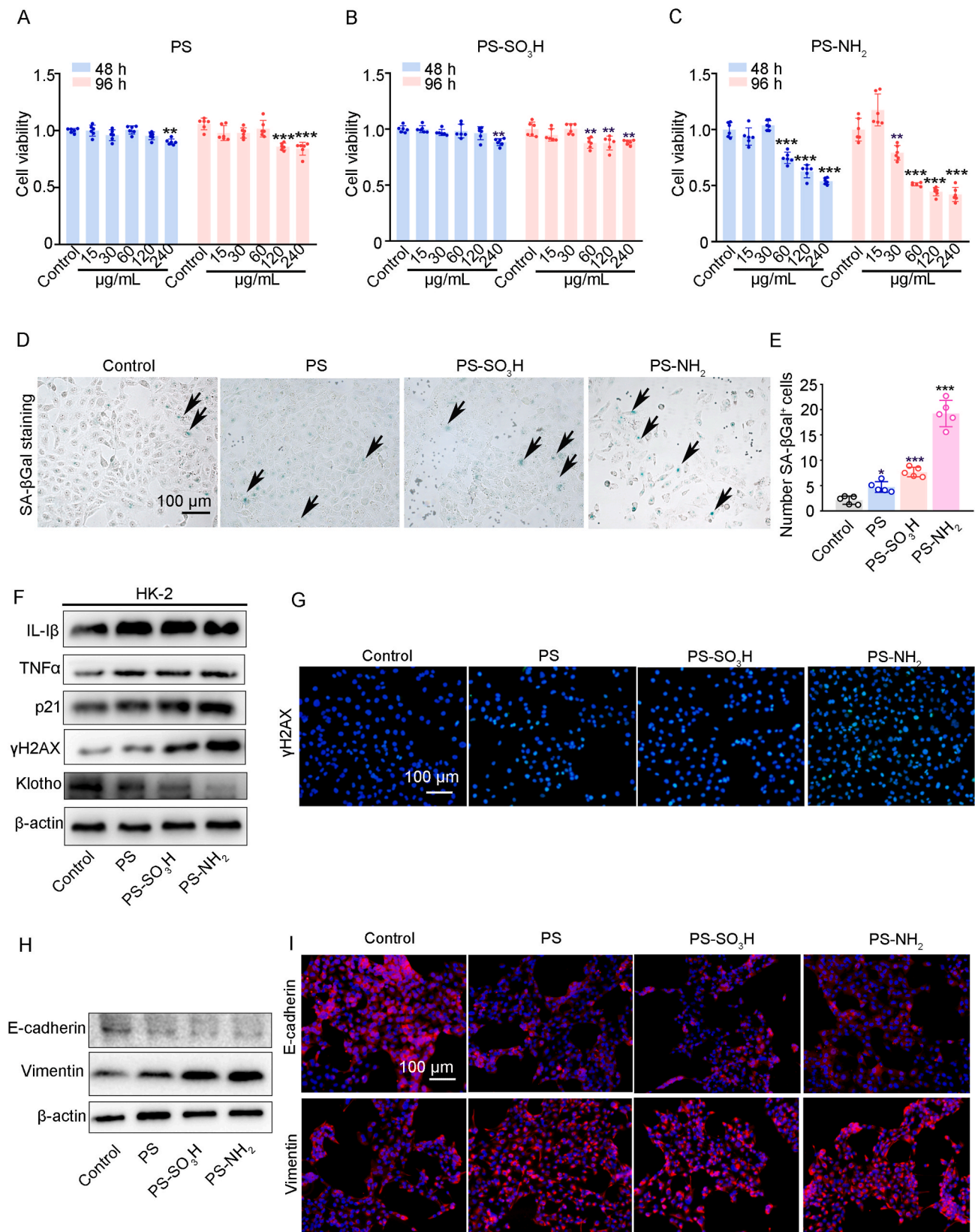


Fig. 4. MPs exposure induced cellular senescence in epithelial cells of renal tubular *in vitro*. The viability of HK-2 cells treated with PS, PS-SO₃H and PS-NH₂ for 48 h and 96 h by MTT assay. (D, E) Representative SA-βGal staining (blue) in HK-2 cells and the SA-βGal⁺ cells were calculated by Image J software (n = 5). (F) The expression levels of IL-1β, TNFα, p21, γH2AX, and Klotho in HK-2 cells were examined by western blotting. (G) Expression levels of γH2AX in HK-2 cells were detected by immunofluorescence assay (n = 5). (H, I) The levels of E-cadherin and vimentin in HK-2 cells were determined by western blotting and immunofluorescence assay. Data were presented as means ± SD. P* < 0.05, P** < 0.01, P*** < 0.001. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

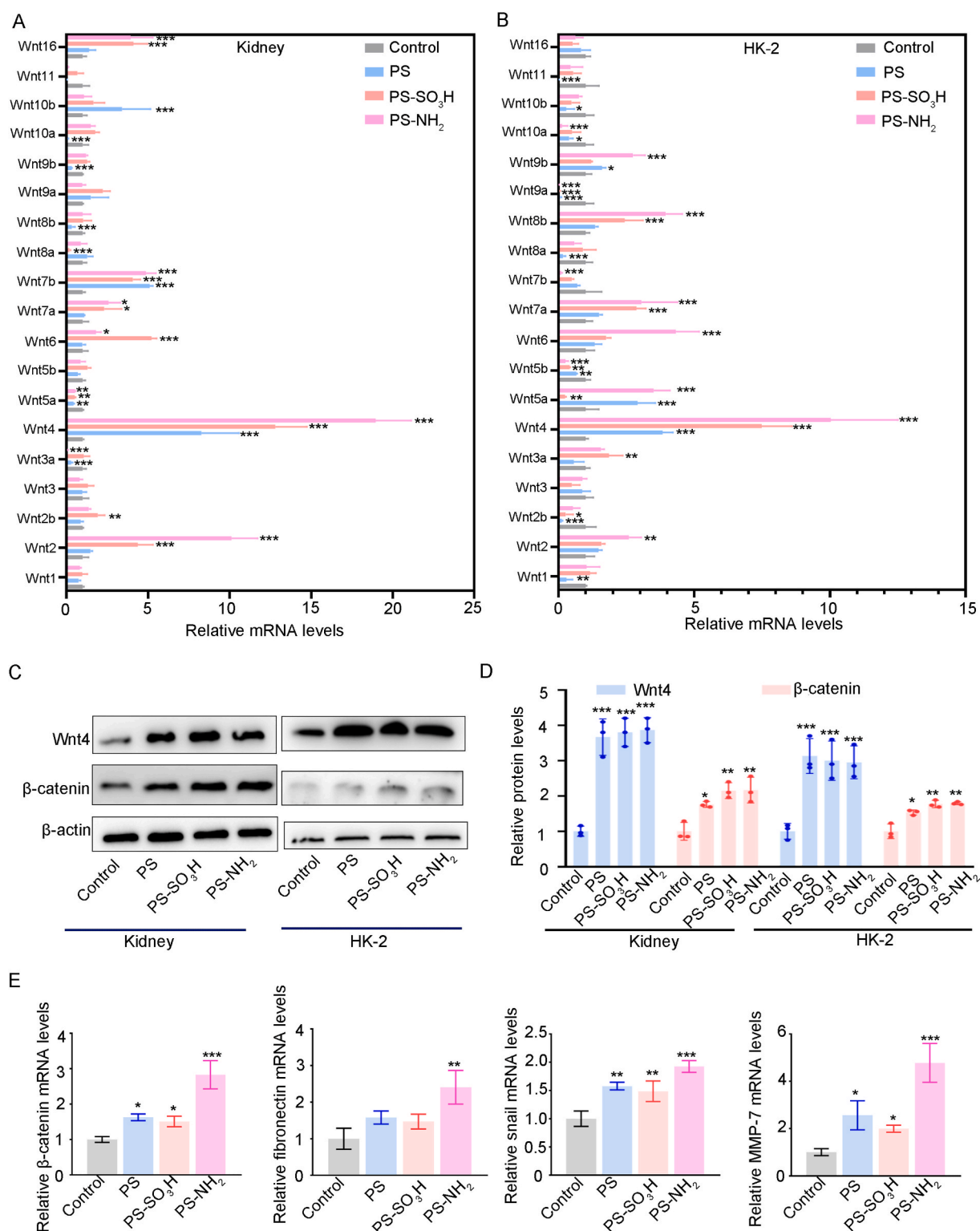


Fig. 5. Identification of the responsible Wnt in senescent epithelial cells in renal tubular. (A, B) QPCR analysis exhibited the relative mRNA levels of Wnt genes in MPs-treated kidney (A) and HK-2 cells (B). (C, D) The protein levels of Wnt4 and β-catenin in kidney and HK-2 cells were examined with western blotting (n = 3). (E) The mRNA levels of Snail 1, β-catenin, fibronectin, and MMP-7 were calculated by quantitative PCR analysis (n = 3). Data were presented as means ± SD. P* < 0.05, P** < 0.01, P*** < 0.001.

mouse model involving MPs exposure during puberty, and the results suggested that MPs induced increased levels of white blood cells, protein, and microalbumin in the urine accompanied by elevated levels of UREA, BUN, CREA and UA, indicating that inflammatory and structure injury may occur in the kidney. Moreover, we observed renal fibrosis in

MPs-exposed mice by Masson and Sirius red staining, which may be the main cause of elevated urinary protein and abnormal renal routine indicators.

Chronic kidney diseases (CKD) are mainly manifested by premature renal aging syndrome, including loss of Klotho, persistent renal

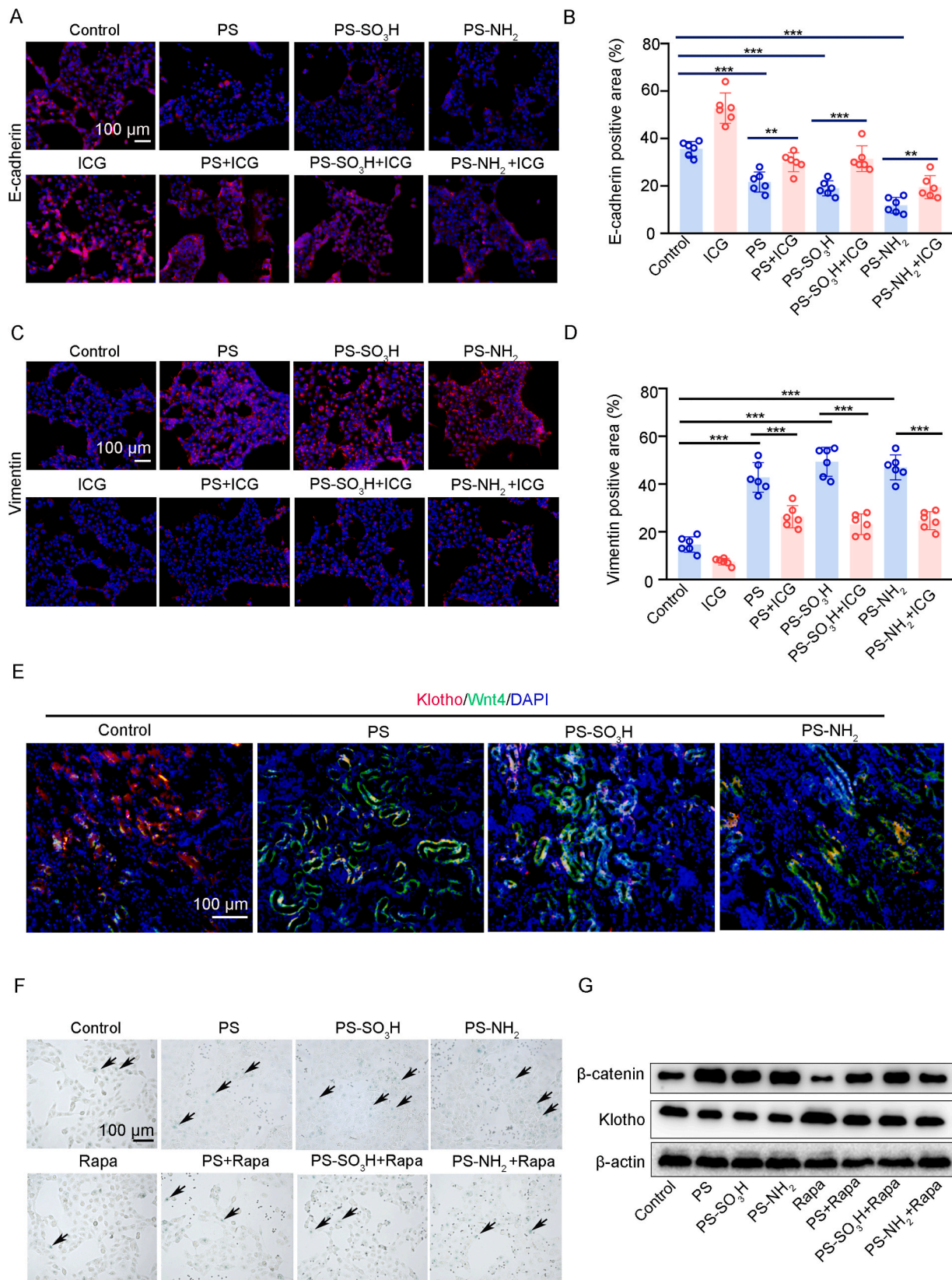


Fig. 6. Klotho mediated Wnt/ β -catenin signaling in senescent renal tubular epithelial cells. (A–D) The levels of E-cadherin and vimentin in HK-2 cells were determined by immunofluorescence assay, and calculated by Image J software ($n = 6$). (E) The levels of Wnt4 (green) and Klotho (red) in HK-2 cells were detected with immunofluorescence assay. (F) Representative SA- β Gal staining (blue) in HK-2 cells was used to examine the senescent cells. (G) The expression of Klotho and β -catenin in HK-2 cells were determined using western blotting. Data were presented as means $\pm$ SD. $P^{**} < 0.01$, $P^{***} < 0.001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

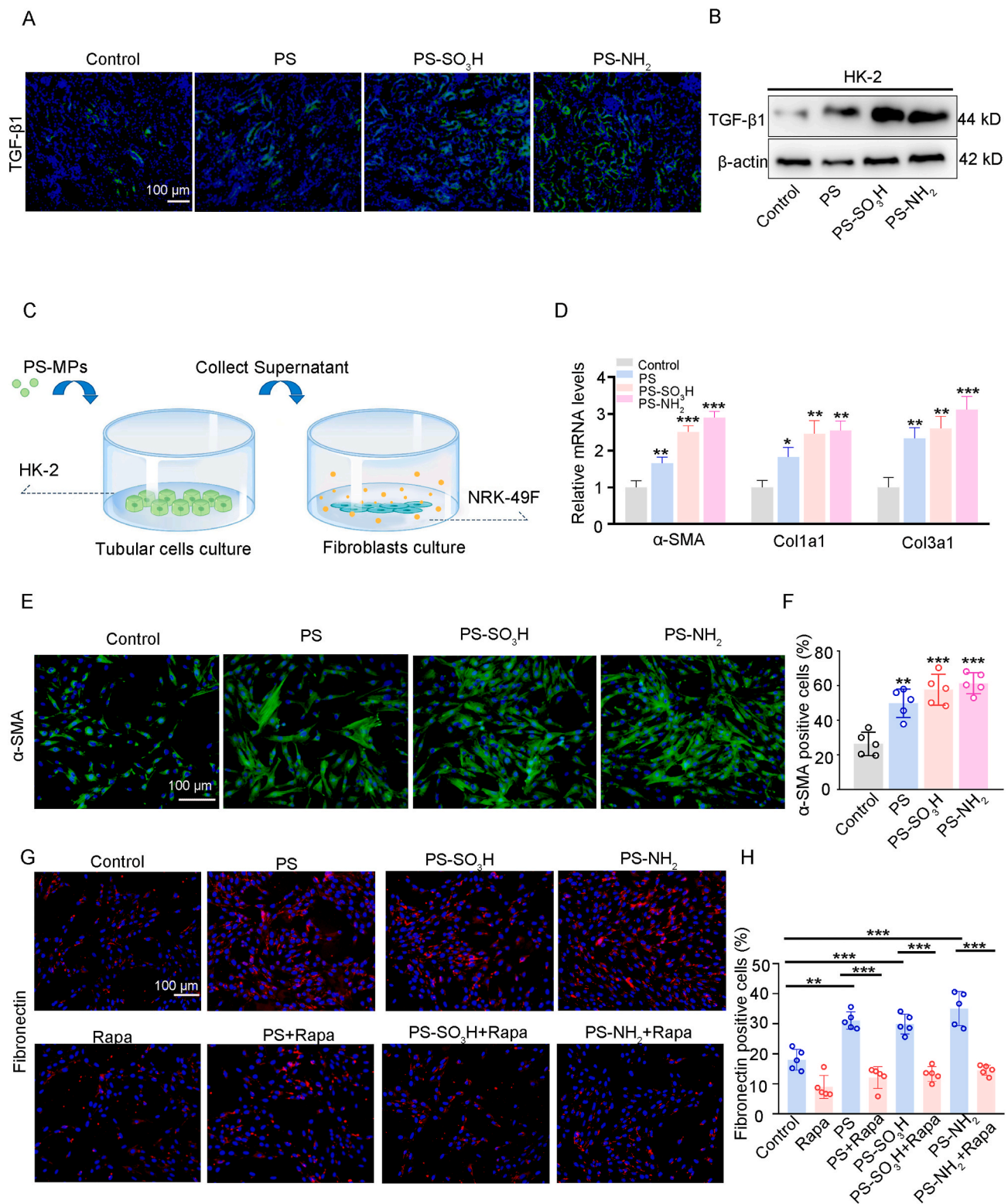


Fig. 7. Senescent epithelial cells in renal tubular secreted TGF-β1 to activate fibroblasts. (A) Expression levels of TGF-β1 in the kidney were determined by immunofluorescence assay, and calculated by Image J software (n = 6). (B) The protein levels of TGF-β1 in HK-2 cells were determined by western blotting. (C–F) HK-2 cells were pretreated with PS, PS-SO₃H and PS-NH₂ MPs for 96 h to induce cellular senescence. The supernatant was collected and used to stimulate NRK-49F cells. (C). The mRNA levels of fibrosis-related genes in NRK-49F cells were calculated using qPCR analysis (n = 3) (D). The levels of α-SMA in NRK-49F cells were determined using immunofluorescence assay, and the α-SMA positive cells were calculated by Image J software (n = 5) (E, F). (G) HK-2 cells were pre-treated with rapamycin (Rapa, 50 nM) for 1 h, and then treated with MPs for 96 h. Then, NRK-49F cells were incubated with the supernatant collecting from the HK-2 cells. The expression levels of fibronectin in HK-2 cells were detected using immunofluorescence assay. Data were presented as means ± SD. P* < 0.05, P** < 0.01, P*** < 0.001.

inflammation, and increased oxidative stress (Buchanan et al., 2020; Munoz-Espin et al., 2013; Zhou et al., 2013). It is worth noting that the accelerated aging phenotype of CKD was observed in specific cell types, mainly tubular cells (Kakoki et al., 2006; Verzola et al., 2008). Cellular senescence is generally considered to be the irreversible fate of damaged cells characterized by cell growth arrest, increased SA- β Gal activity, and enhanced expression of senescence-associated proteins including p21, p16^{INK4A}, p53 and p19 (Baker et al., 2016; Luo et al., 2018). We are the first to support the evidence MPs induced senescence of tubular cells *in vivo* (mice) and *in vitro* (HK-2 cells). We found elevated activities of SA- β Gal and the loss of anti-aging protein Klotho in epithelial cells of renal tubular and HK-2 cells. Senescent epithelial cells in renal tubular not only lose the ability to proliferate, but also secrete senescence associated secretory phenotype with enhanced secretion of proinflammatory cytokines including TNF α and IL-1 β to attract inflammatory cells and exert adverse effects on surrounding cells. Similarly, other studies have also demonstrated that MPs-treated renal cells exhibited increased levels of mitochondrial ROS and inflammation-related proteins (Goodman et al., 2022; Wang et al., 2021). *In vitro*, we observed that MPs-treated HK-2 cells exhibited cell cycle arrest, increased number of senescent cells, and enhanced secretion of inflammatory factors, consistent with the results *in vivo*.

Renal fibrosis can be significantly alleviated by the clearance of senescent cells (Zhu et al., 2015). However, the potential mechanism of MPs promoting renal cell senescence is unclear. Klotho has anti-aging properties, and the content of Klotho in serum decreases with human aging and aging-related diseases, such as hypertension, cardiovascular disease, and chronic kidney disease (Buchanan et al., 2020). Therefore, the regulation of Klotho activity may be considered an effective target for the treatment of senescence. In this study, our results suggested that MPs exposure induced the suppression of Klotho *in vivo* and *in vitro*, and rapamycin (Klotho agonist) exerted inhibitory effects on cellular senescence, indicating that Klotho mediated cellular senescence in renal fibrosis.

In healthy kidneys, the Wnt/ β -catenin pathway is silent, but the continuous activation of this signal is closely related to the development of renal fibrosis (von Toerne et al., 2009). Notably, cellular senescence exerts a vital role in the pathological processes of these diseases (Braun et al., 2012). In our study, we observed that Wnt4 exerts a decisive part in driving renal fibrosis. Despite Wnt4 expression being almost undetectable in normal mouse kidneys, Wnt4 expression was significantly induced and mainly distributed in epithelial cells in renal fibrosis tissues of mice with MPs treatment. We also observed that the levels of Wnt4 are closely associated with the expression of Klotho, accompanied by the increase of γ H2AX foci, which is a marker of renal tubular injury. The above results suggest that Wnt4 activation is closely related to cellular senescence during renal injury. Several studies have shown that the anti-aging protein Klotho mediates the activation of the Wnt pathway, such as Wnt7b, Wnt9a, and Wnt10a (Chen et al., 2019; Gu et al., 2019), suggesting that the decrease in Klotho expression caused by MPs may be an important reason for Wnt4 activation. Our *in vitro* study further indicated that Wnt4 expression in HK-2 cells induced β -catenin signaling activation and accelerated EMT in epithelial cells of renal tubular. At the same time, the expression of EMT-related proteins fibronectin, MMP-7 and snail increased significantly, indicating MPs can promote EMT in epithelial cells of renal tubular. Recent studies have suggested that EMT of epithelial cells in renal tubular can cause cell cycle arrest and secrete a large number of pro-fibrotic cytokines, which is an important promoting factor leading to renal fibrosis (Lovisa et al., 2015). HK-2 cells were preprocessed with ICG-001 effectively alleviated the occurrence of EMT in cells. To support our data, it has been previously reported that the Wnt/ β -catenin pathway can promote EMT in cancer cells and bone marrow mesenchymal stem cells (Tang et al., 2020).

Another important feature of senescent cells is the secretion of large amounts of pro-inflammatory and pro-fibrotic cytokines (SASP phenotype) (Pan et al., 2023), and we found that MPs exposure can promote

the senescence of epithelial cells in renal tubular and secrete senescence-related pro-fibrotic cytokines TGF- β 1. Proliferation and activation of interstitial fibroblasts is a vital pathological feature of chronic renal fibrosis. A large number of studies have indicated that stimulating NRK-49F cells by collecting HK-2 conditional media can lead to the occurrence of renal fibrosis (Luo et al., 2018; Overstreet et al., 2015; Zhang et al., 2023). In our study, we found that conditional medium derived from epithelial cells in the renal tubular with MPs treatment can markedly promote the activation of renal fibroblasts. Previous reports have supported our findings that TGF- β 1 secreted by epithelial cells in renal tubular mediates the proliferation and activation of renal fibroblasts (Meng et al., 2015).

5. Conclusions

In summary, MPs-induced abnormal expression of Klotho exerts a vital role in mediating renal tubular senescence in renal fibrosis. We have confirmed that the persistent up-regulation of Wnt4 caused by MPs can induce EMT in senescent epithelial cells of renal tubular and inhibit the proliferation and repair of normal epithelial cells. Moreover, the decrease in Klotho levels caused by MPs may be an important reason for Wnt4 activation. At the same time, TGF- β 1 secreted by senescent epithelial cells in renal tubular induced the proliferation and activation of renal fibroblasts, and accelerated the transformation of damaged kidneys into fibrosis. The above results indicated that targeted inhibition of cellular senescence may be a new strategy to alleviate renal fibrosis caused by MPs.

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CRediT authorship contribution statement

Chun Pan: Formal analysis, Data curation. **Xinglong Wang:** Investigation, Funding acquisition, Conceptualization. **Zhencheng Fan:** Writing – review & editing. **Wenwen Mao:** Resources, Project administration. **Yujie Shi:** Validation, Supervision. **Yin Wu:** Validation, Conceptualization. **Tingting Liu:** Writing – review & editing, Software. **Zhuobin Xu:** Supervision, Software. **Huihui Wang:** Writing – original draft, Software, Conceptualization. **Hao Chen:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fct.2024.114888>.

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